LAB 1: EXERCISES

Shell Programming on Linux

Advanced Programming Techniques (ELT 3296)
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Exercise 1: File and directory management shell commands

Objective: Practice basic shell commands for file and directory manipulation.
Requirements: Execute the following requests using shell commands:

1. Move to your user directory.
2. Inside your user directory, create two subdirectories `Slides`, `Labs` and a file named `Solution.txt`.
3. Go to the `Labs` directory.
4. Inside the `Labs` directory, create a subdirectory named `Lab1` and a file named `Ex1.txt`.
5. Copy the file `Ex1.txt` from the `Labs` directory to the `Lab1` directory.
6. Remove the file `Ex1.txt` inside the `Labs` directory.
7. Rename the file `Ex1.txt` to `Exercise1.txt`.
8. Add write mode access to the file `Exercise1.txt` for other users.
9. Remove the directory `Slides`.

Exercise 2: File System Navigation

Requirements: Continue from Exercise 1, go to the `Lab1` directory and try this sequence of commands and record the results of these commands.

1. `cd`
2. `pwd`
3. `ls -al`
4. `cd .`

5. `pwd`
6. `cd ..`
7. `pwd`
8. `ls -al`
9. `cd ..`
10. `pwd`
11. `ls -al`

Exercise 3: Creating and Writing to Files

Requirements: Create a file `Hello.txt` inside the `Lab1` directory. Then using the `echo` command and redirection to write the below contents to the file `Hello.txt`.

```
Hello everyone!
This is Advanced Programming Techniques course.
There are many exercises for you in this course.
Let's practice what we learn in this course with exercises.
Are you ready?
```

Exercise 4: Process management shell commands

Requirements: Use the I/O redirection and pipe mechanisms and Unix commands to perform the following tasks:

1. Count the number of words "course" appeared in the file `Hello.txt` (you created in Exercise 3) and save results to a file `Ex4_results.txt`.
2. Print the 3rd line of the file `Hello.txt`, then count the number of words in this line and save results to the end of the file `Ex4_results.txt`.
   (*Hint: using* `sed`*,* `wc` *and pipe mechanism*)
3. Store the names and attributes of the directories and files inside your user directory.
4. Count the total number of regular files and directories inside your user directory and save it to the end of the file `Ex4_results.txt`.
5. Count the total number of directories inside your user directory and save it to the end of the file `Ex4_results.txt`.
6. Count the total number of processes currently in the system and save results to the end of the file `Ex4_results.txt`.

Exercise 5: Basic Shell Script

Requirements: Write a set of commands in a shell script file named `greeting.sh` that does the following:

1. Contains a comment with your name, the name of this script, and the purpose of this script.
2. Displays the string "Hello + username".
3. Displays the name and version of this operating system.
4. Displays the group ID (GID) of the current user.
5. Displays the current date and time.
6. Displays a list of all the files in the current user's directory.
7. Displays the values of the `TERM`, `PATH`, and `HOME` variables.
8. Displays "Goodbye + current time" to the user.

Exercise 6: Command Line Parameters

Requirements: Write a shell script named `test.sh` (with any number of parameters and parameter list) that displays the following information:

- Program name
- Number of command line parameters
- First command line parameter entered by the user
- All command line parameters entered by the user

Exercise 7: Number Base Conversion

Requirements: Write a shell script to convert a decimal integer to hexadecimal. The shell script takes a command line parameter which is a decimal integer to convert.
Hint: using the `bc` *command to convert from decimal to hexadecimal.*

Exercise 8: Fibonacci Sequence Script

Requirements: Program a shell script to print the Fibonacci sequence up to the n-th number. The Fibonacci sequence has the following characteristics:

- $F(0) = 0$
- $F(1) = 1$

- $F(n) = F(n-1) + F(n-2)$ with $n > 1$

Specific conditions:

- Enter the value n from the user.
- Print the Fibonacci sequence from $F(0)$ to $F(n)$.